

Why Did We Plot the Cluster Sizes and Then Fit an Exponential Decay?

1) Visual Inspection of the Distribution

First, we plotted the average number of feeding events (y_i) versus the cluster size (x_i , i.e., the number of pellets in each feeding event) to:

- Inspect how frequently certain cluster sizes occur (identify any visible pattern).
- Verify if the counts fall off quickly or slowly as cluster size increases.

This preliminary visualization helps confirm whether the data might follow a known pattern such as an exponential decay, a power law, or other distributions.

2) Motivation for an Exponential Decay Model

In many biological or behavioral contexts, larger events become increasingly rare in a way that can be described by exponential decay. An exponential decay function can be written as:

$$y(x) = a e^{-bx},$$

where:

- a = scale parameter (the theoretical y -intercept),
- b = decay rate (how quickly the exponential decreases),
- x = the independent variable (cluster size in our case).

In this setting,

x = cluster size (number of pellets), $y(x)$ = average number of feeding events of that size.

By fitting an exponential decay, we test the hypothesis that the frequency of larger cluster sizes diminishes at an exponential rate.

3) Fitting the Model and Computing Parameters

To find the parameters a and b , we typically solve a nonlinear least squares problem of the form:

$$\min_{a,b} \sum_i \left(y_i - a e^{-bx_i} \right)^2,$$

where (x_i, y_i) are the observed data points (cluster size vs. average count). Once we obtain $\hat{a}$ and $\hat{b}$, the fitted curve is

$$\hat{y}(x) = \hat{a} e^{-\hat{b}x}.$$

4) Goodness of Fit: R^2

To quantify how well this exponential model explains the data, we compute the coefficient of determination R^2 :

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

where

y_i = observed data value (average count),

$\hat{y}_i = \hat{a} e^{-\hat{b}x_i}$ (model prediction),

$\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$ (mean of the observed values).

An R^2 value closer to 1 indicates that the model explains most of the variance in the data.

5) Interpretation

By first visualizing the raw distribution, we confirm that the data appear to follow an exponential-like drop-off in feeding event counts for larger cluster sizes. Then, using the exponential decay model:

$$y(x) = a e^{-bx},$$

we quantify:

- How quickly larger cluster sizes become rare (via b).
- The initial scale of feeding event counts (via a).
- The overall suitability of the exponential model (via R^2).

Thus, combining a clear visual inspection with an appropriate mathematical model (exponential decay) allows us to characterize how the frequency of feeding events changes as the cluster size increases.